

beam-mom-offset.cc

BEAM momentum offset applied to Target

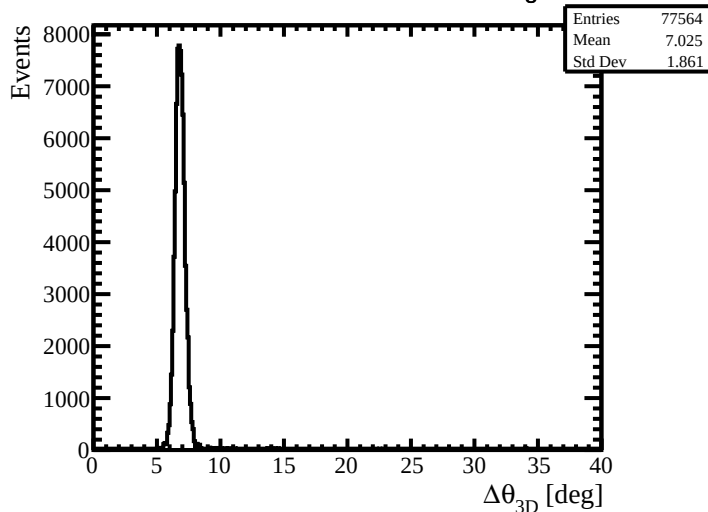
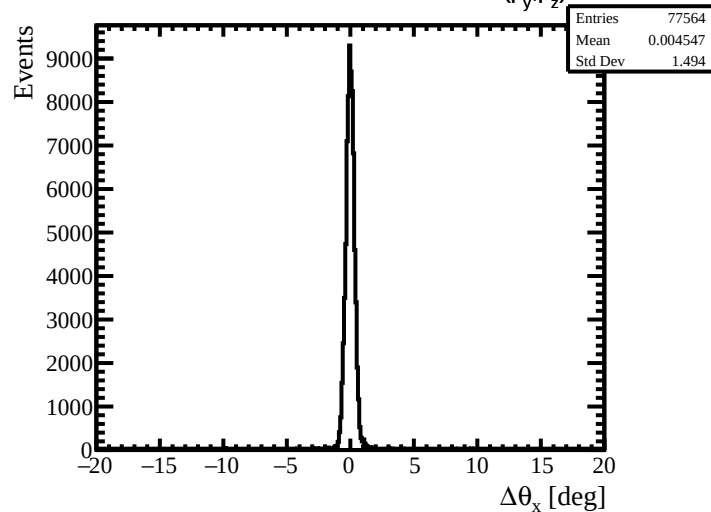
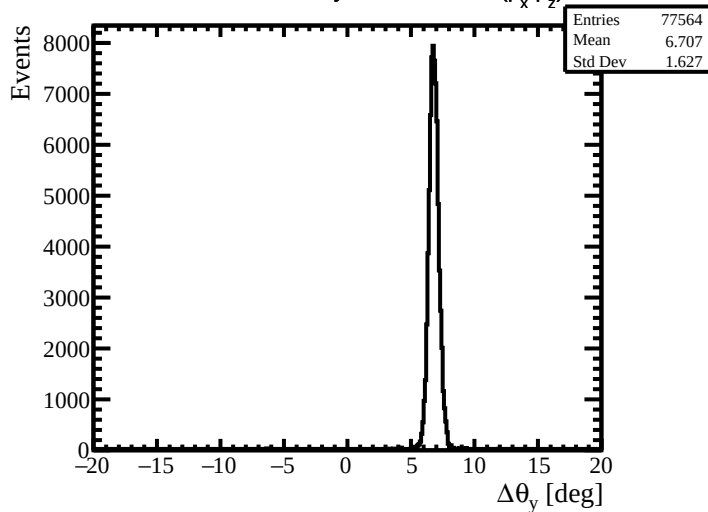
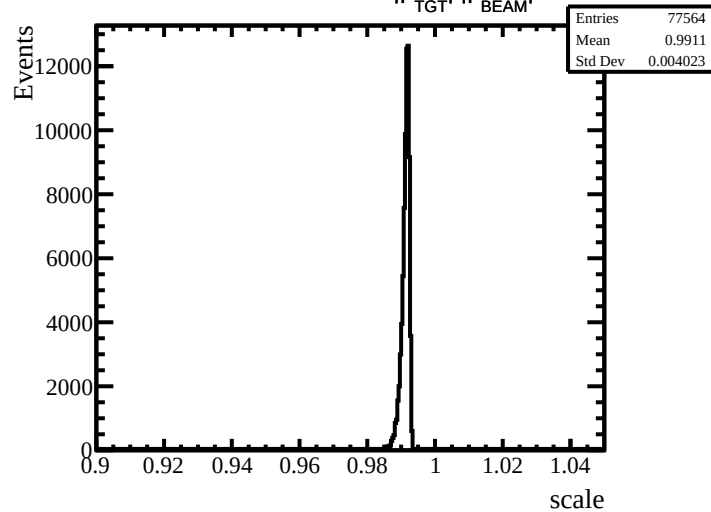
Input: /gpfs/group/had/sks/Users/haein/simul-data/e72-beam/e72_kbeam_735.root

rot x/y/z = 0.00455 / 6.70687 / 35.53579 deg

momentum scale = 0.99112402

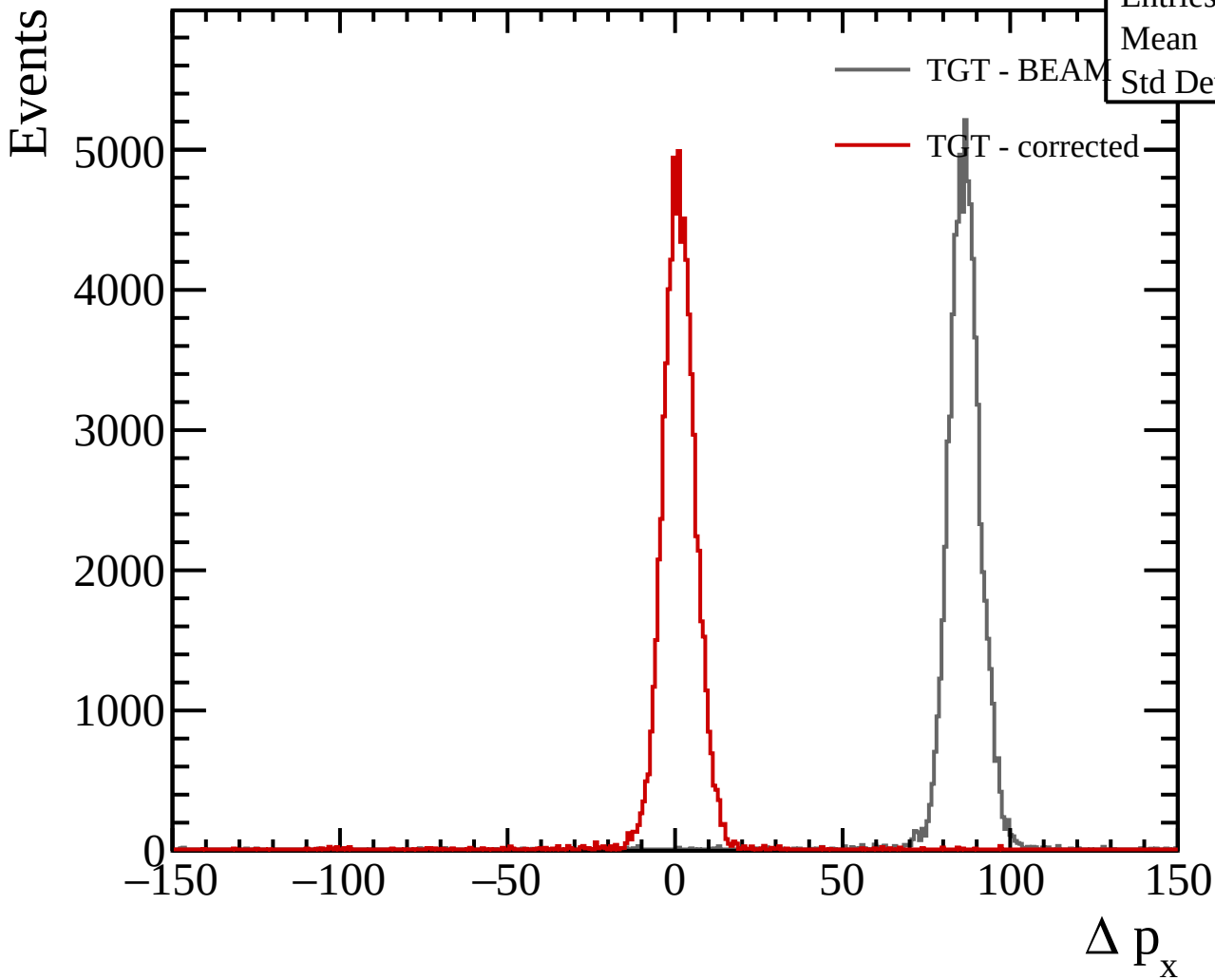
Generated at: 2026-07-17 14:55:14

BEAM to TGT 3D direction change

Rotation around x axis: $\Delta \text{atan2}(\mathbf{p}_y, \mathbf{p}_z)$ Rotation around y axis: $\Delta \text{atan2}(\mathbf{p}_x, \mathbf{p}_z)$ Momentum scale $|\mathbf{p}_{\text{TGT}}|/|\mathbf{p}_{\text{BEAM}}|$ 

TGT - BEAM p_x

Entries	77564
Mean	84.89
Std Dev	16.17



TGT - BEAM p_y

Events

3500

3000

2500

2000

1500

1000

500

0

-80

-60

-40

-20

0

20

40

60

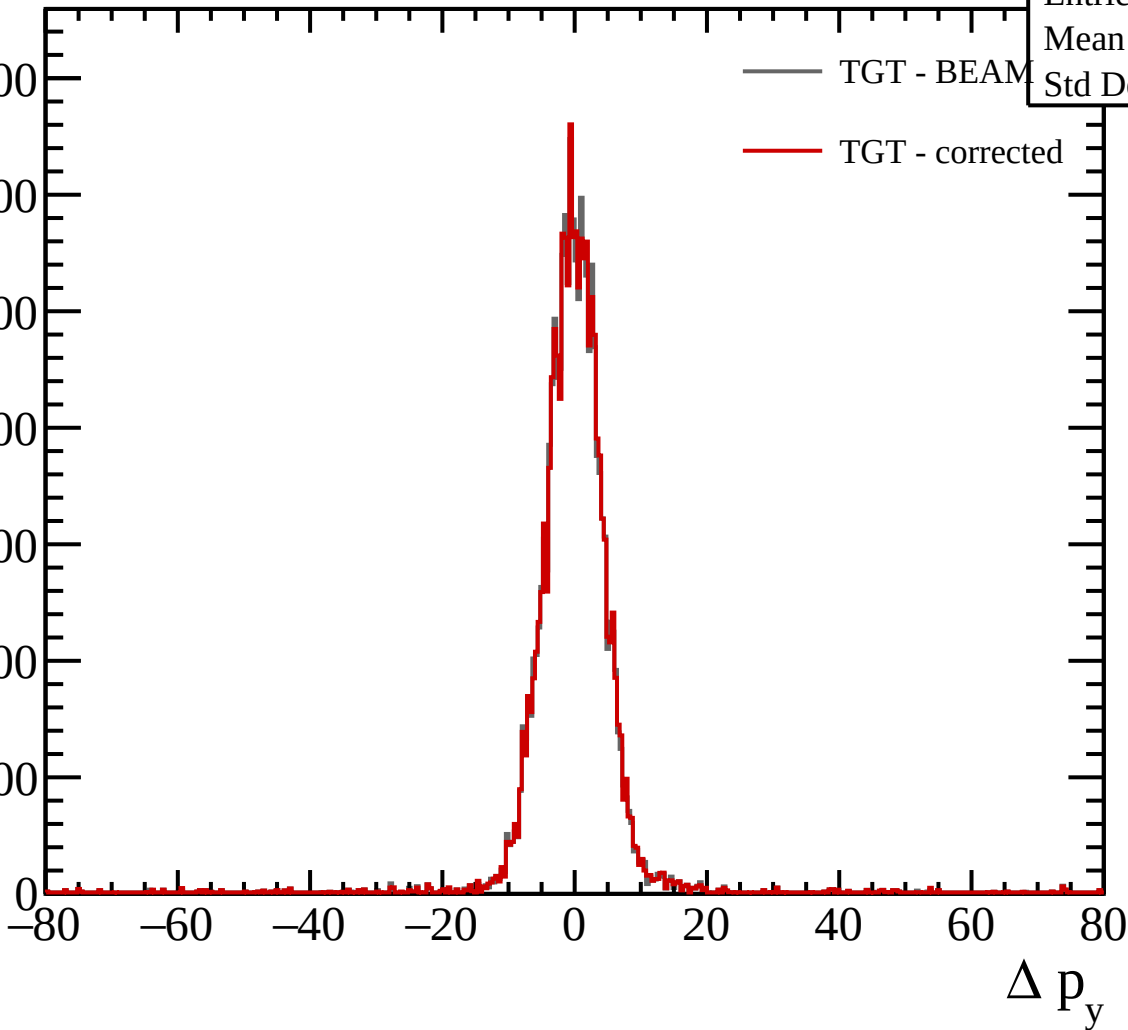
80

Δp_y

— TGT - BEAM

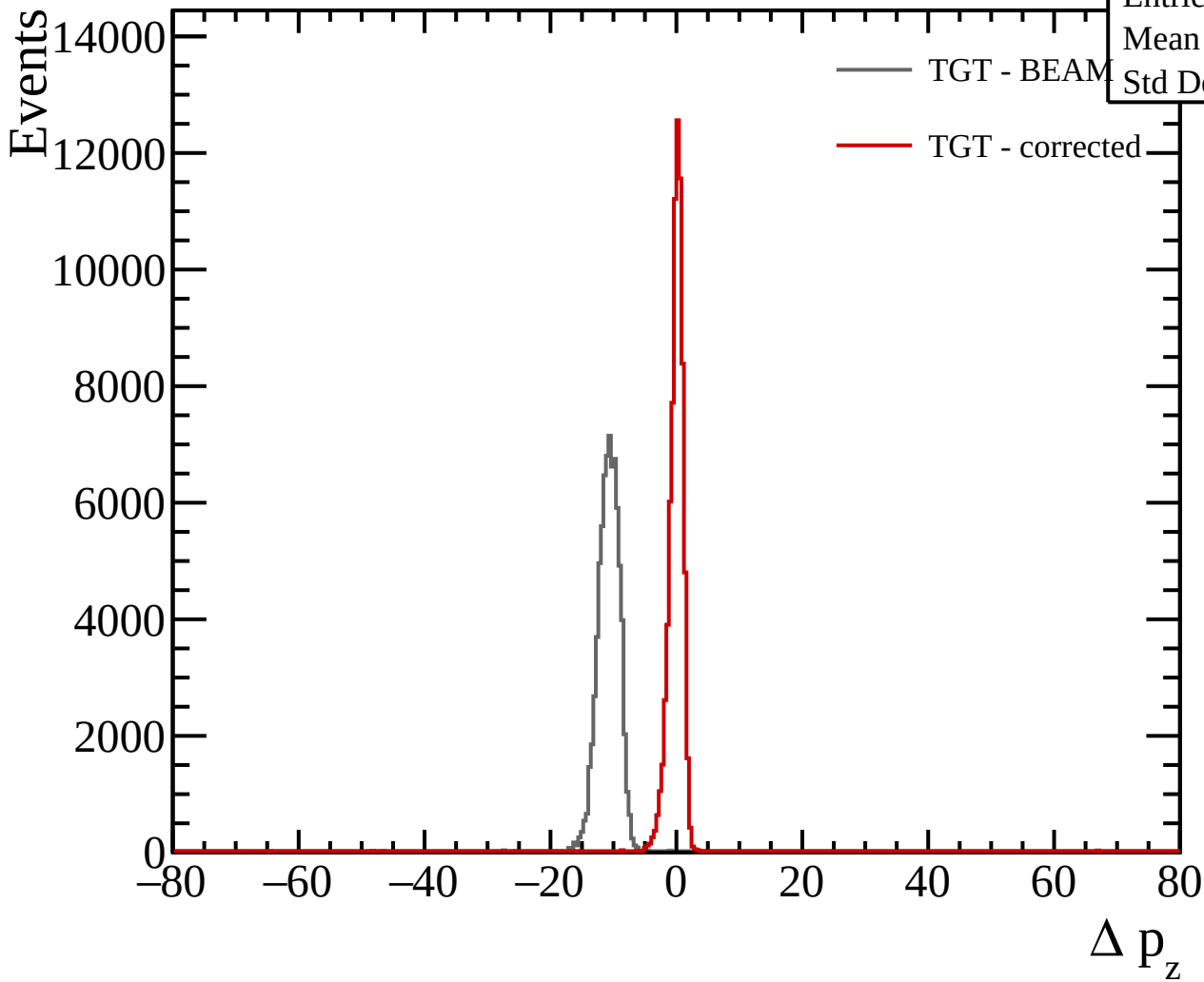
— TGT - corrected

Entries	77564
Mean	-0.1162
Std Dev	7.543

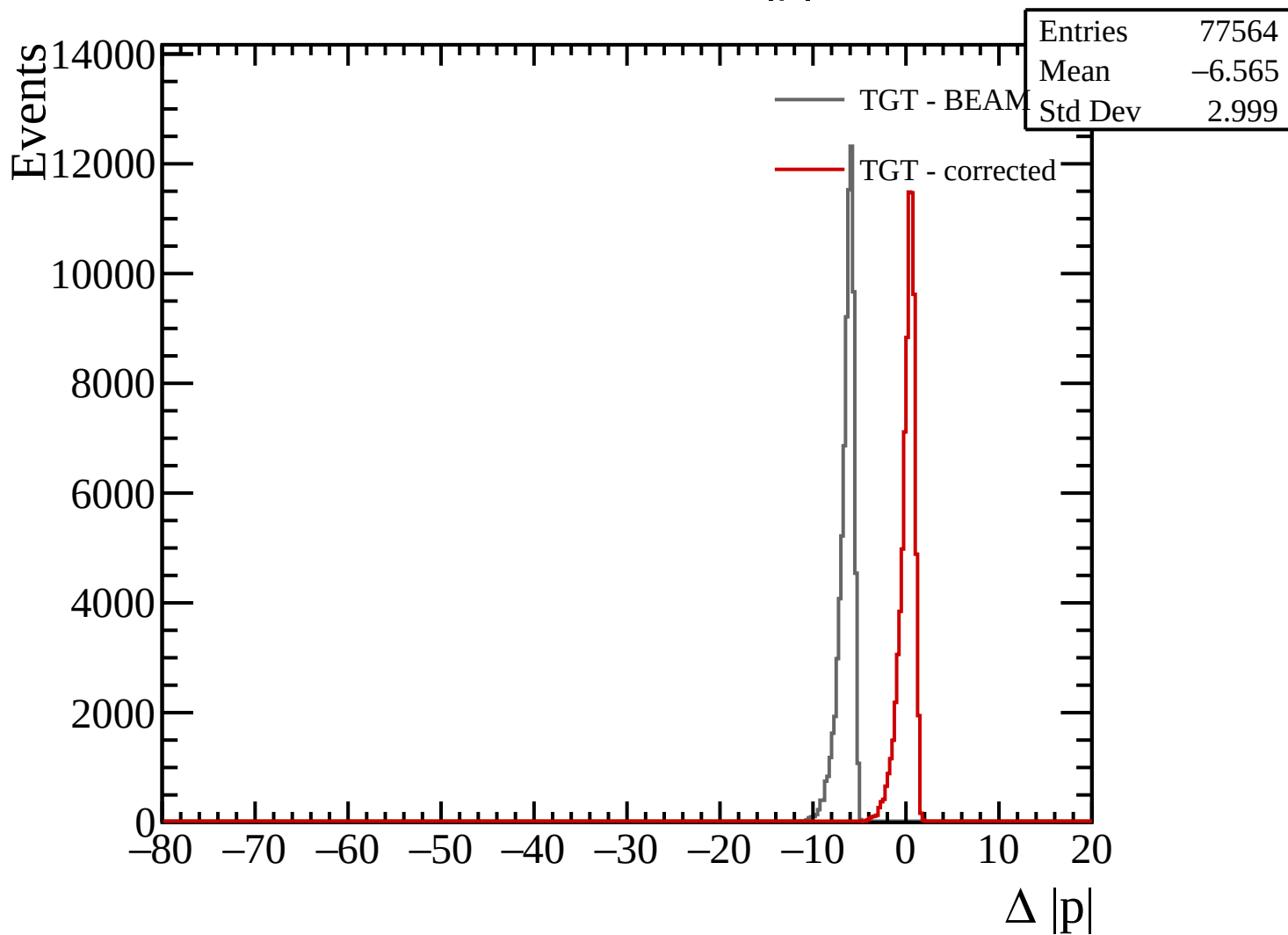


TGT - BEAM p_z

Entries	77564
Mean	-10.74
Std Dev	4.546



TGT - BEAM |p|



Angle between TGT and BEAM

Events

7000

6000

5000

4000

3000

2000

1000

0

0

2

4

6

8

10

12

14

16

18

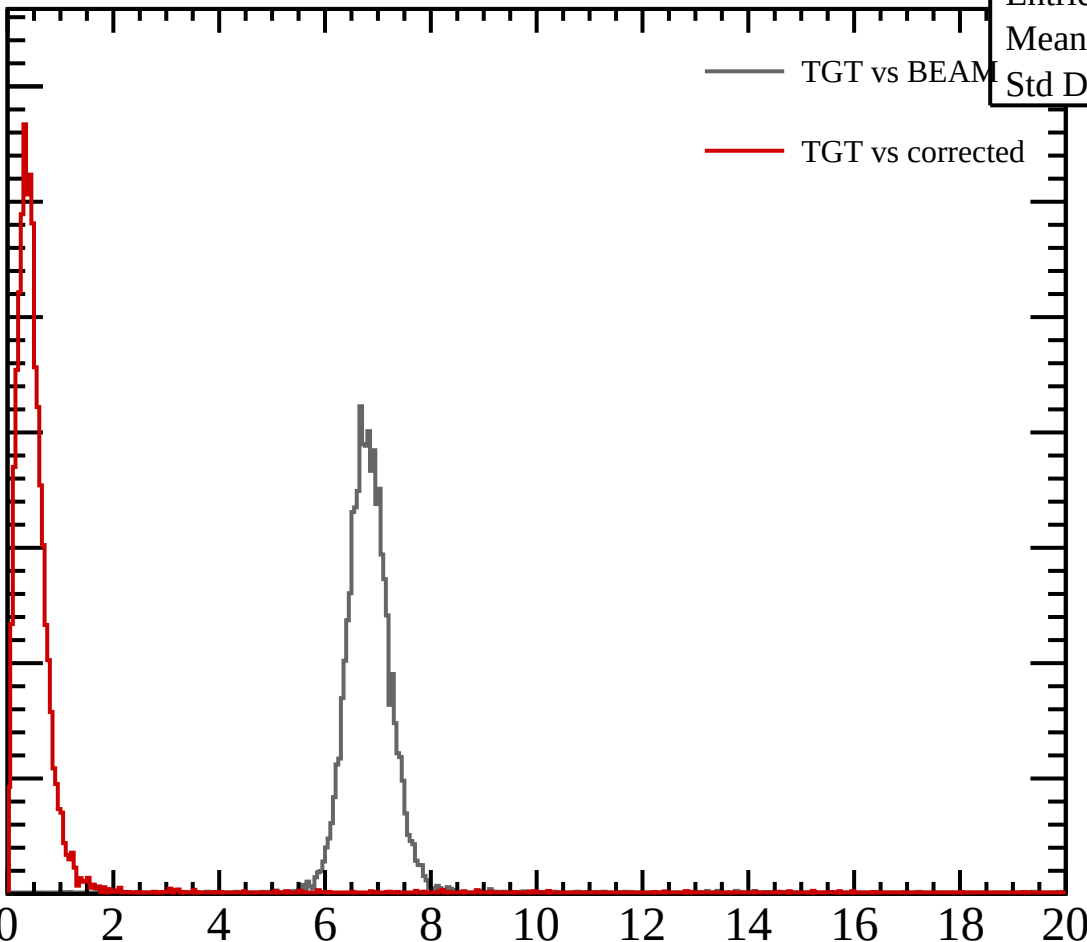
20

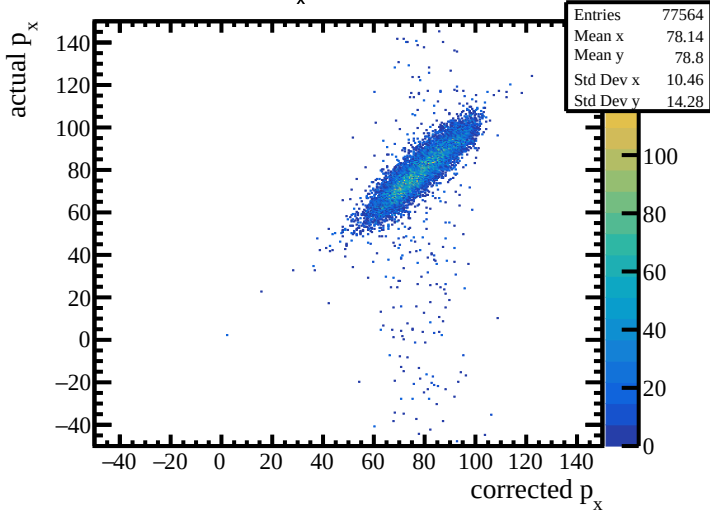
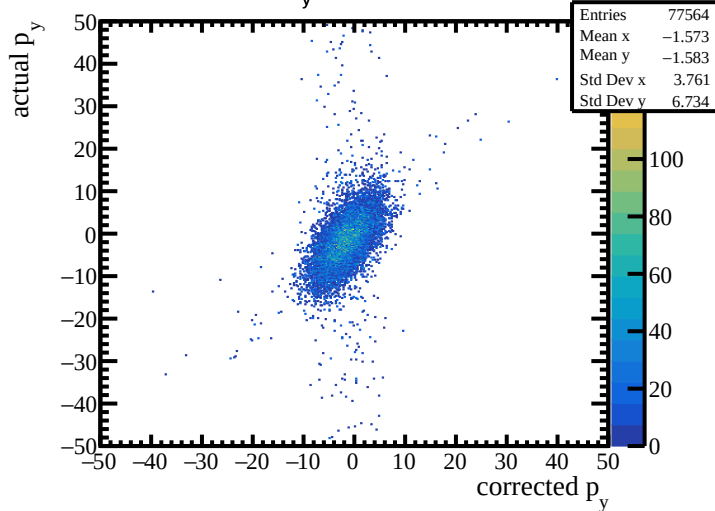
angle [deg]

— TGT vs BEAM

— TGT vs corrected

Entries	77564
Mean	6.904
Std Dev	0.9518



Target p_x : actual vs correctedTarget p_y : actual vs correctedTarget p_z : actual vs corrected